

Control Structures

LO2

23-1-26

Aim – Use loops and mathematical concepts to produce a star pattern and factorial of entered number

Theory – It uses for loops in a fixed range to produce the desired output. The range function starts from 0 to the given number hence in the factorial program, we had to set the start to 1 as any number multiplied by zero is zero. In the star generation program, we used for loops to generate the empty spaces and the *'s wherever required according to the pattern

A1.

n = 5

```
for i in range(n):  
    for j in range(n - i - 1):  
        print(' ', end="")  
    for j in range(2 * i + 1):  
        print('*', end="")  
    print()
```

```
for i in range(n - 2, -1, -1):  
    for j in range(n - i - 1):  
        print(' ', end="")  
    for j in range(2 * i + 1):  
        print('*', end="")  
    print()
```



A2.

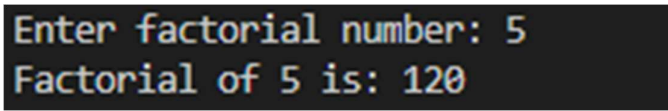
```
count = input("Enter factorial number: ")
```

```
factorial = 1
```

```
for i in range(1, int(count) + 1):
```

```
    factorial = factorial * i
```

```
print("Factorial of " + str(count) + " is: " + str(factorial))
```



```
Enter factorial number: 5  
Factorial of 5 is: 120
```

A screenshot of a terminal window with a dark background. The first line shows the prompt 'Enter factorial number:' followed by the user input '5'. The second line shows the program output 'Factorial of 5 is: 120'.

Conclusion

It demonstrates the use of loops in python and how efficiently apply mathematical concepts and patterns in python.